Executive Benchmark

Evidence cutoff: Jun 30, 2025 | Independent public-data reconstruction

Comparability guardrail: ecosystem roles are not interchangeable. Scalability scores are ordinal and lens-dependent.

company_name | ecosystem_role | evidence_status | scheme

35	33	100.0%	57M	250K	PASS
Sources	Primary sources	QA pass rate	Waymo rider-only miles	Paid trips / week	Model health

company_name	ecosystem_role
Mobileye	OEM technology supplier
NVIDIA	Compute and software platform prov
Tesla	Vehicle and consumer ADAS develop
Waymo	Vehicle or mobility operator

company_name	Data collection	Deployment	Sensor approach	Simulation	T
Mobileye	5.0	5.0	4.0	4.0	
NVIDIA	3.0	4.0	4.0	5.0	
Tesla	5.0	4.0	4.0	3.0	
Waymo	4.0	5.0	5.0	5.0	
Total	**4.3**	**4.5**	**4.3**	**4.3**	

Weighted Scalability Score | Ordinal, lens-dependent



Deployment Company Label	metric_name	metric_value	unit
California program total	California AV test miles total	9068861	mile/ye
California program total	California AV test miles with safety driver	3945171	mile/ye
California program total	California fully autonomous test miles	552895	mile/ye
Multiple companies	Consumer vehicle deliveries	1789226	vehicle,
Multiple companies	Cumulative vehicles with Mobileye technology	200000000	vehicle
Multiple companies	EyeQ and SuperVision systems shipped	29000000	system,
Multiple companies	Fully autonomous miles	50000000	mile
California program total			

company_name	capability_category	evidence_status	limitation
Mobileye	Data collection	Verified	Installed vehicles span different f
Mobileye	Deployment	Verified	Shipments are mostly ADAS com
Mobileye	Sensor approach	Verified	Product tiers differ substantially
Mobileye	Simulation	Moderately supported	Limited comparable public metri
Mobileye	Training infrastructure	Verified	Public information does not discl
NVIDIA	Data collection	Verified	No unified exposure metric
NVIDIA	Deployment	Verified	Partner announcements and proc
NVIDIA	Sensor approach	Verified	Final sensor configuration is set b

Technology & Workflow

Evidence cutoff: Jun 30, 2025 | Independent public-data reconstruction

Technology evidence is role-specific; supplier, platform, operator, and consumer-ADAS measures are not equivalent.

company_name | technology_category | relationship_type | evidence_status

Role-aware autonomous-driving workflow

Collect → Select → Label → Train → Simulate → Validate → Deploy → Monitor

Operator, platform, supplier, and consumer-ADAS roles contribute at different stages; presence does not imply equivalence.

company_name	Camera perception	Closed-loop simulation	Cloud training platform	Fleet learni
Mobileye	1		1	
NVIDIA		1	1	
Tesla	1		1	
Waymo	1	1	1	
Total	3	2	4	

company_name	sensor_strategy_summary	comparability_boundary
Mobileye	Camera-first with optional radar/lidar redundancy	Very broad installed base across r
NVIDIA	Reference multimodal sensor architecture	Indirect deployment through OEI
Tesla	Primarily camera/vision	Broad consumer fleet; system ren
Waymo	Camera, lidar, radar, external audio	Rider-only operations with define

company_name	in_vehicle_compute_summary	ecosystem_r
Mobileye	EyeQ family	OEM techno
NVIDIA	DRIVE AGX/Orin/Thor roadmap	Compute an
Tesla	Tesla FSD computer	Vehicle and
Waymo	Waymo Driver custom compute stack	Vehicle or m

company_name	capability_category	evidence_status	source_id	limitation
Mobileye	Data collection	Verified	SRC023	Installed vehicles span different features and generatic
Mobileye	Deployment	Verified	SRC023	Shipments are mostly ADAS components, not Level 4 v
Mobileye	Sensor approach	Verified	SRC023	Product tiers differ substantially
Mobileye	Simulation	Moderately supported	SRC023	Limited comparable public metrics
Mobileye	Training infrastructure	Verified	SRC023	Public information does not disclose training cluster sc
NVIDIA	Data collection	Verified	SRC018	No unified exposure metric
NVIDIA	Deployment	Verified	SRC020	Partner announcements and production volumes are n
NVIDIA	Sensor approach	Verified	SRC019	Final sensor configuration is set by customer programs

Technology & Workflow

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GPU Cost & Scalabilitv

Evidence cutoff: Jun 30, 2025 | Independent public-data reconstruction

Scenario costs are standardized analytical workload outputs, not company expenditures. Scalability scores are ordinal and lens-dependent.

scenario_name | cloud_provider | gpu_model | instance_type | discount_model | scheme

$84M Total scenario cost

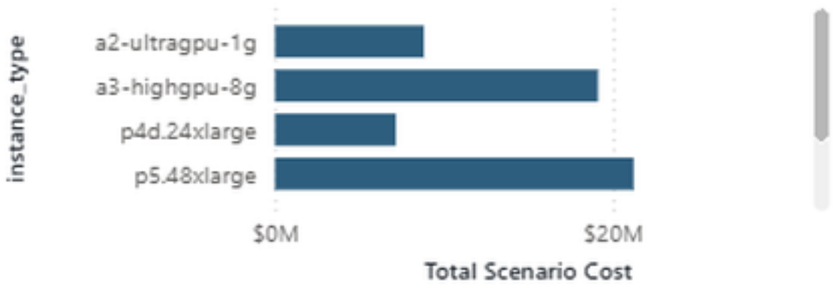
$70M Training compute

$14M Simulation compute

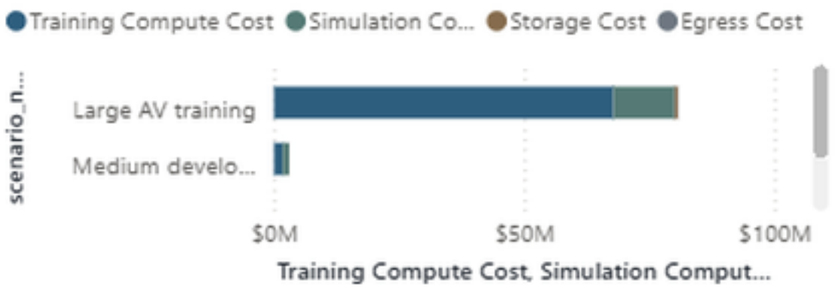
$653K Storage

$90K Egress

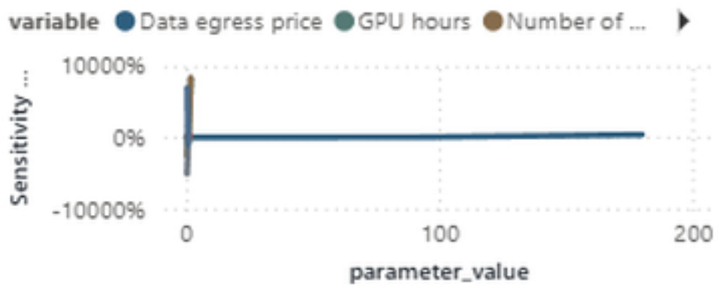
Scenario Cost by Provider and Instance



Cost Composition



Sensitivity to Scenario Inputs



criterion_name	Mobileye	NVIDIA	Tesla	Waymo	Total
Commercial flexibility	5.00	5.00	2.00	3.00	3.75
Cost scalability	5.00	4.00	5.00	2.00	4.00
Data scale	5.00	3.00	5.00	4.00	4.25
Deployment maturity	5.00	4.00	4.00	5.00	4.50
Geographic scalability	5.00	5.00	5.00	2.00	4.25
Hardware dependence	4.00	2.00	4.00	3.00	3.25
In-vehicle compute	4.00	5.00	4.00	4.00	4.25
OEM or fleet integration	5.00	5.00	2.00	3.00	3.75
Total	**4.50**	**4.17**	**3.75**	**3.42**	**3.96**

Weighted Score by Decision Lens | Ordinal

